

Recovery of configured and observed insulin action profiles from routine automated insulin delivery records: a cross-sectional analysis of 31 users

Tim Street

Diabettech. Correspondence: tim@diabettech.com

Aims. Automated insulin delivery (AID) systems represent insulin using an assumed activity profile parameterised by a time to peak action and a duration of action. These parameters are set by convention and are not verified against the individual. This study assessed whether both the profile a system is configured with, and the profile a person's glucose appears to follow, can be recovered from routinely collected records.

Methods. Records from 31 adults using oref-derived AID systems (9 AndroidAPS, 22 Trio) were analysed. The configured profile was recovered by deconvolving each system's logged insulin-on-board series against its delivered doses under the exponential insulin model; this relationship is an algebraic identity and therefore admits an exact positive control. The observed profile was estimated non-parametrically as a finite impulse response on five-minute glucose increments, with a second-difference smoothness penalty, a non-negativity constraint, and a time-of-day drift profile pooled across days. Both estimators were validated against simulated data with known parameters, and the non-parametric estimator additionally against simulated closed-loop feedback.

Results. Configured profiles were recovered with relative residuals of 0.03 to 0.13 in the 21 participants whose dose records reconciled with their logged insulin-on-board, and recovered peaks clustered on manufacturer preset values. An identifiable peak was obtained in 29 of 31 participants; in the remainder the recovered response had no mode and no peak is reported. Observed peaks had a median of 42 min (range 28–112) against configured values of 55 min (35–77); the correlation between the two was +0.03. Duration of action was not identifiable in 10 of 31 participants. No dependence of the observed peak on dose size was detected across a twenty-two-fold range of dose.

Conclusions. The configured profile is recoverable exactly and provides a practical audit of what a system is computing. The observed profile is recoverable with useful precision but is systematically shorter than both the configured value and published clamp pharmacodynamics, and the configured value carries little information about it. These are observational estimates obtained under conditions that differ materially from those of clamp studies, and are not evidence that any setting is incorrect.

Keywords: automated insulin delivery; insulin pharmacodynamics; deconvolution; continuous glucose monitoring; type 1 diabetes

1. Introduction

Automated insulin delivery systems predict the effect of insulin already delivered. That prediction rests on an assumed activity profile, parameterised by a time to peak action t_p and a duration of action t_d . In oref-derived systems these are drawn from a small set of preset values, typically 55 or 75 minutes for the peak, and applied uniformly to all users of a given insulin analogue.

Interindividual variation in subcutaneous insulin absorption is well described, and the product information for rapid-acting analogues states explicitly that the duration of action varies with dose, injection site, blood flow, temperature and physical activity [9, 11]. Whether an individual's own records can characterise their profile is therefore of practical interest, but the data present two obstacles. A closed-loop system delivers a near-continuous sequence of small reactive doses rather than isolated impulses, so conventional impulse-response averaging has almost no material to work with: in one participant, of approximately 19,800 boluses only 234 were separated by a clear hour and 38 by two hours. Glucose additionally responds to carbohydrate, endogenous production, physical activity and sensor error, and the largest of these confounders — unannounced carbohydrate — is by definition absent from the record.

Two questions are distinguished here that are commonly conflated. First, what profile is a system configured with? This is an audit question with an exact answer, because the system's own logged insulin-on-board is a deterministic function of its dose history and that profile. Second, what profile does the glucose response appear

to follow? This is an inferential question with an observational answer. The objectives were to establish whether each is recoverable from routine records, to quantify the agreement between them, and to identify sources of bias in the estimates.

2. Methods

2.1 Data

Records were obtained from 31 adults using oreof-derived AID systems and uploading to Nightscout. Each record comprises per-cycle controller decisions at five-minute cadence — sensor glucose, carbohydrate on board, insulin on board and its basal component, and step counts where the uploader provides them — together with a treatment stream of delivered insulin and entered carbohydrate. Observation periods ranged from 16 to 370 days. No intervention was made and no protocol imposed.

The delivery system is not recorded in the extract and was determined from three independent features: whether boluses carry a type field, whether bolus insulin-on-board is uploaded or must be derived as total minus basal, and whether step counts are present. All 31 classifications were concordant across the three features, giving 9 AndroidAPS and 22 Trio users. No demographic data were available.

Insulin the user reports having given outside the pump is uploaded under a distinct event type carrying an amount but no insulin type. Such records are present in 14 of 31 participants and are bimodal in size: mostly a few units, consistent with a pen dose of the same analogue, but with a minority of 20 to 35 U at approximately twelve-hourly spacing, which is not a plausible single rapid-acting bolus. Because leverage in a linear model scales with the square of the regressor, these few records exert influence far beyond their share of delivered insulin — 37.8% of dosing leverage from 11.1% of delivered units in the most affected participant.

The two estimators require opposite treatment of these records. The controller credits this insulin into its own insulin-on-board essentially in full (0.99 U of insulin-on-board per unit logged, measured across the events themselves), so it forms part of the identity the configured-profile estimator rests on and is retained there; withholding it raises the relative residual of the most affected participant from 0.18 to 0.60. The observed-profile estimator asks instead how the insulin behaves in glucose, where an unknown and possibly long-acting preparation represented by a rapid-acting kernel would bias the estimate late. These records are therefore excluded from the observed-profile estimator, and the sensitivity to that choice is reported in section 3.6.

2.2 Insulin model

Both estimators refer to the exponential model used by oreof-derived systems [1]. For peak time t_p and duration t_d in minutes,

$$\tau = t_p(1 - t_p/t_d) / (1 - 2t_p/t_d), \quad a = 2\tau/t_d, \quad S = [1 - a + (1 + a)e^{-t_d/\tau}]^{-1}$$

The fraction of a unit dose remaining active at time t , and its rate of action, are

$$f(t) = 1 - S(1 - a)[(t^2/(\tau t_d(1-a)) - t/\tau - 1)e^{-t/\tau} + 1]$$

$$A(t) = (S/\tau^2) \cdot t \cdot (1 - t/t_d) \cdot e^{-t/\tau}$$

both zero outside $[0, t_d]$. The parameterisation places the maximum of A exactly at t_p : substituting τ from (1) into the derivative of (3) gives $A'(t_p) = 0$ identically. This was verified numerically over $t_p \in [25, 120]$ and $t_d \in [180, 1440]$, with agreement to 0.02 min, and $A(t)$ was confirmed equal to $-df/dt$ to 2×10^{-10} .

The published pharmacodynamics underlying these curves derive from euglycaemic-clamp studies reported in the product information; the two-parameter functional form used to interpolate them is a community derivation without a peer-reviewed source [1]. Table 1 sets the label values against the presets the systems adopt.

Analogue	Onset	Maximum effect	Duration	Peak assumed by system
insulin lispro [8]	~15 min	not stated	2–5 h	75 min
insulin aspart [9]	10–20 min	1–3 h	3–5 h	75 min
insulin glulisine [10]	10–20 min	not stated	~4 h	75 min
faster-acting insulin aspart [11]	5 min earlier than aspart	1–3 h	not stated	55 min
ultra-rapid lispro [12]	20 min	1–3 h	5 h	55 min

Table 1. Pharmacodynamic parameters as stated in European product information, against the peak assumed by oref-derived systems. Every label stating a maximum effect gives the same interval irrespective of analogue class; two state no peak. The 55 and 75 minute presets are conventions rather than label values.

2.3 Recovery of the configured profile

A system's logged bolus insulin-on-board is a deterministic function of its dose history and configured profile:

$$\text{IOB}(t) = \sum_j d_j \cdot f(t - t_j; t_p, t_d)$$

Doses were binned onto the controller's five-minute grid and (4) evaluated as a discrete convolution with a kernel of $\lceil t_d/5 \rceil + 1$ taps. Parameters were estimated by bounded least squares (trust-region reflective) over $t_p \in [10, 180]$ and $t_d \in [120, 1440]$ min. Observations within ten minutes after any bolus were excluded: insulin-on-board steps by the whole dose at delivery, so sub-grid error in the timing of that step produces a residual of up to one dose, and the bin in which the step appears differs between systems. Exclusion reduced the relative residual from 0.11–0.32 to 0.02–0.09 and removed a downward bias on t_d ; in the one participant whose configured duration was independently known to be 600 min, the estimate moved from 314 to 487 min. The peak estimate changed by less than 0.2 min.

Identifiability of t_d was assessed by leverage rather than by interval width. Holding $\hat{t}_p$ fixed, the predicted series was recomputed at $t_d \in \{240, 360, 480, 600, 900, 1440\}$ and compared with the fitted series; where no alternative shifted the prediction by more than twice the fit residual, the duration was reported as not identifiable. This criterion was adopted because bootstrap intervals proved unreliable for this parameter: in the participant with a known duration of 600 min, a 28-day fit returned 314 min with a 95% interval of [260, 378], the narrowest interval obtained and one excluding the true value.

2.4 Recovery of the observed profile

The observed response was estimated without assuming a functional form. Glucose increments were regressed on lagged doses:

$$\Delta g_t = -\sum_{k=0}^K \beta_k d_{t-k} + c_{\text{clock}(t)} + u_{\text{day}(t)} + \varepsilon_t$$

β_k , one free coefficient per five-minute lag, $k = 0 \dots K$ with $K = 72$ (six hours), constituting the activity profile; c , a time-of-day profile of 48 half-hour clock bins with one dropped for identifiability, shared across all days; u , a per-day intercept.

An earlier formulation fitted a parametric profile within individual overnight fasting windows, each with its own drift term. That design proved under-identified: within a three-to-five-hour window the insulin activity profile is slowly varying and approximately linear in time, as is the dawn rise from which it must be separated, with median $|r|$ between the insulin regressor and elapsed time of 0.73–0.81 at four-hour windows rising to 0.99 at two hours. Across 96 defensible specifications the estimates for individual participants ranged over 40 to 90 minutes, with the single choice of including a linear drift term moving two participants by 23 and 35 minutes. That approach was abandoned and its results are not reported.

Equation (5) resolves this by treating drift as a repeatable time-of-day profile estimated across all days simultaneously. Dawn is locked to clock time; doses are not, and that difference in timing provides the identifying variation. It exists only when days are pooled, and for the same reason the sample comprises every fasting period across the twenty-four-hour cycle rather than overnight windows alone.

Samples were eligible where carbohydrate on board was zero, no carbohydrate had been entered in the preceding three hours, sensor glucose lay between 40 and 350 mg/dL, the sample fell outside any post-rescue window, and — where step data were available — the rolling hourly step count was below 200.

The peak was taken over lags of 15 min and above. No rapid-acting analogue has an onset of action below 10 min by its own product information, so an earlier peak is not physiologically admissible; and reverse causality concentrates in the first bins, since the controller doses in response to rising glucose and the non-negativity constraint cannot represent that as a negative coefficient. In one participant a single-bin coefficient of 2.45 at 5 min outranked a broad, well-supported mode of 2.15 at 65 min. The restriction altered the estimate for that participant alone; every other recovered peak lay at or beyond 20 min.

Adjacent lags are strongly collinear, so the unpenalised estimate is uninformative. A second-difference (Tikhonov) penalty was applied [2], with β constrained non-negative:

$$\hat{\beta} = \arg \min_{\beta \geq 0} \|y - X\beta\|^2 + \lambda \|D\beta\|^2$$

where X stacks the lagged-dose, clock and day blocks and D is the second-difference operator acting on the lag coefficients only. The constrained problem was solved as a bounded least-squares system on the augmented design $[X; \sqrt{\lambda}D]$ against $[y; 0]$. The smoothing weight was selected by generalised cross-validation [3],

$$\text{GCV}(\lambda) = n \cdot \text{RSS}(\lambda) / [n - \text{tr } H(\lambda)]^2, \quad H(\lambda) = X(X^\top X + \lambda D^\top D)^{-1} X^\top$$

evaluated over a logarithmic grid using $\text{tr}(XHX^\top) = \text{tr}(HX^\top X)$. Selection was performed on the unconstrained ridge solution, for which the hat matrix is available in closed form, and the selected value applied to the constrained fit; the constraint binds only in the tail. The peak was taken as $5 \cdot \arg \max_k \hat{\beta}_k$, with the corrections described in section 2.6.

2.5 Validation

Equation (4) is an identity, so the parametric estimator has an exact positive control: applied to a system's own logged insulin-on-board it must return the configured parameters, and any departure indicates a defect in the method rather than a finding about insulin.

For the non-parametric estimator a control was constructed by retaining each participant's real dose series, sampling grid and eligibility mask and replacing observed glucose with glucose simulated from a known kernel, plus a clock-locked drift term and white noise calibrated to that participant's residual scale. Recovery was tested against three generating families — the exponential model above, gamma curves of differing shape parameter, and bi-exponentials of differing tail ratio — since an estimator recovering only its own family would be flexible rather than non-parametric. A noise-free condition was included, in which recovery must be exact.

Both controls generate glucose from the dose series and therefore treat the input as exogenous. In closed-loop data it is not: the controller doses in response to the same glucose it regulates. In these records a dose correlated +0.25 to +0.55 with the glucose change in the preceding five minutes and −0.05 to +0.04 with the following thirty. A further control simulated the loop closed, with glucose responding to insulin through a known kernel while a proportional controller dosed on the recent rise, at gains spanning an open loop to an aggressively reactive one. Performance was assessed against the open-loop case rather than an absolute tolerance, since an imprecise estimator errs at zero gain also [5].

Four properties of the extract were verified independently of the kernel: agreement between uploaded and derived bolus insulin-on-board where both were available; absence of duplicate treatment records; equality of the logged insulin-on-board step at an isolated bolus with the recorded dose; and alignment, assessed by correlating the dose series against the insulin-on-board increment in each bin and by refitting with the dose series displaced by −30 to +30 min.

2.6 Bias corrections

Two systematic biases in the estimator are corrected in all reported values. First, the target is a forward difference regressed on d_{t-k} , so the change across an interval is attributed to a lag of $5k$ min when the mean lag

across that interval is $5k + 2.5$; an alignment term of $+2.5$ min is added at reporting. Second, glucose increments are autocorrelated (lag-one residual correlation $\approx +0.44$), violating the independence assumed by (7). Thinning the sample moved estimates later and converged at one-in-three (fifteen-minute spacing), by 0 to $+15$ min depending on participant; one-in-three thinning is therefore used throughout. The alignment term is applied at reporting rather than within the estimator, because the simulated controls are generated on the same discrete grid and carry no corresponding offset.

2.7 Identifiability of the peak

The estimator returns a coefficient vector, and every coefficient vector has a maximum. Quoting that maximum as a peak presumes the curve has a mode, which is a property of the estimate and not something the method guarantees. Two conditions were therefore required before a peak was reported, both computed on the same lag range as the peak search. *Concentration* is the fraction of kernel mass lying within ± 30 min of the maximum, and *prominence* is $(\max - \text{median of the positive coefficients})/\max$. Thresholds were fixed from the model already in use rather than from these data: the slowest profile any of these systems permits (peak 75 min, duration 480 min) scores 0.34 and 0.53, and a fast one (35/480) scores 0.58 and 0.93, so floors of 0.25 and 0.40 lie below every admissible curve while still rejecting one with no mode. Participants failing either condition are reported as having no identifiable peak and are excluded from summary statistics over peaks; their argmax is recorded in the per-participant output but is not carried forward.

The second condition addresses a failure the first cannot see. An oscillating estimate can place ample mass near its argmax and still be reporting noise, and whether it oscillates is governed by λ , which cross-validation selects rather than the data fix. Each fit was therefore repeated at one tenth and ten times the selected λ , and a peak was reported only where the three agreed to within 15 min. Across the cohort the median spread over that hundred-fold range was 5 min, and one participant exceeded the tolerance at 25 min: a kernel oscillating between adjacent lags on the smallest sample in the series, whose argmax lay between two substantially smaller neighbours. That participant's peak is withheld on the same basis.

2.8 Statistical analysis

Intervals were obtained by block bootstrap resampling whole days with replacement and refitting [4]; days rather than observations were the resampling unit because consecutive five-minute samples are strongly dependent. Agreement between configured and observed peaks is summarised by the Pearson correlation and by the distribution of differences. No hypothesis test is reported: the analysis is descriptive, the sample is one of convenience, and no adjustment for multiplicity would render the comparisons confirmatory. Analyses were performed in Python 3.14 with NumPy, SciPy and pandas.

3. Results

3.1 Validation

All positive controls were passed (0 failures). Table 2 reports recovery of known peaks by the non-parametric estimator. Recovery was exact under the noise-free condition for all three generating families, including the two the estimator does not assume.

Generating curve	True peak (min)	Recovered, noise-free	Recovered, realistic noise
AAPS exponential	35	35	40
AAPS exponential	55	55	55
AAPS exponential	75	75	85
gamma, shape 3 (wrong family)	45	45	50
gamma, shape 6 (wrong family)	35	35	35
bi-exponential, ratio 8 (wrong family)	45	45	45

Table 2. Recovery of a known activity peak from simulated glucose using real dose series and sampling.

Under simulated closed-loop feedback, mean absolute error did not increase with controller gain relative to the open-loop reference (excess -3.3 min). Induced endogeneity reached +0.27 to +0.37, within the range observed in the records, indicating the control exercised the intended failure mode.

Extract integrity was confirmed for all participants: treatment records were free of duplicates; the insulin-on-board step at an isolated bolus matched the recorded dose to within the decay expected across the bin; step offsets were confined to the zero or one-bin quantisation expected of five-minute sampling; and where bolus insulin-on-board was uploaded directly it agreed with the derived value to 0.00025 U. Displacing the dose series confirmed that zero shift minimised the residual.

3.2 Configured profiles

Configured peaks were recovered with 95% intervals spanning 1 to 2 min and relative residuals of 0.028 to 0.126 in the 21 participants whose records reconciled. Estimates were stable across disjoint halves of the observation period in all but one participant. Recovered values clustered on manufacturer presets (Figure 1).

In 10 participants the relative residual exceeded 0.15 on a relationship that is an identity, indicating that the dose record does not correspond to what the system counted. Candidate explanations include extended or multiwave boluses, partial delivery, and timestamps offset from the delivery registered by the controller. Their configured profiles are reported but should be regarded as provisional.

Refitting on disjoint halves identified a change of configured profile in participant P8 (83 to 52 min) that no treatment record documented. Rolling seven-day windows localised the change to a single week. The profile-change records present for this participant were dated one week after the kinetics changed.

3.3 Observed profiles

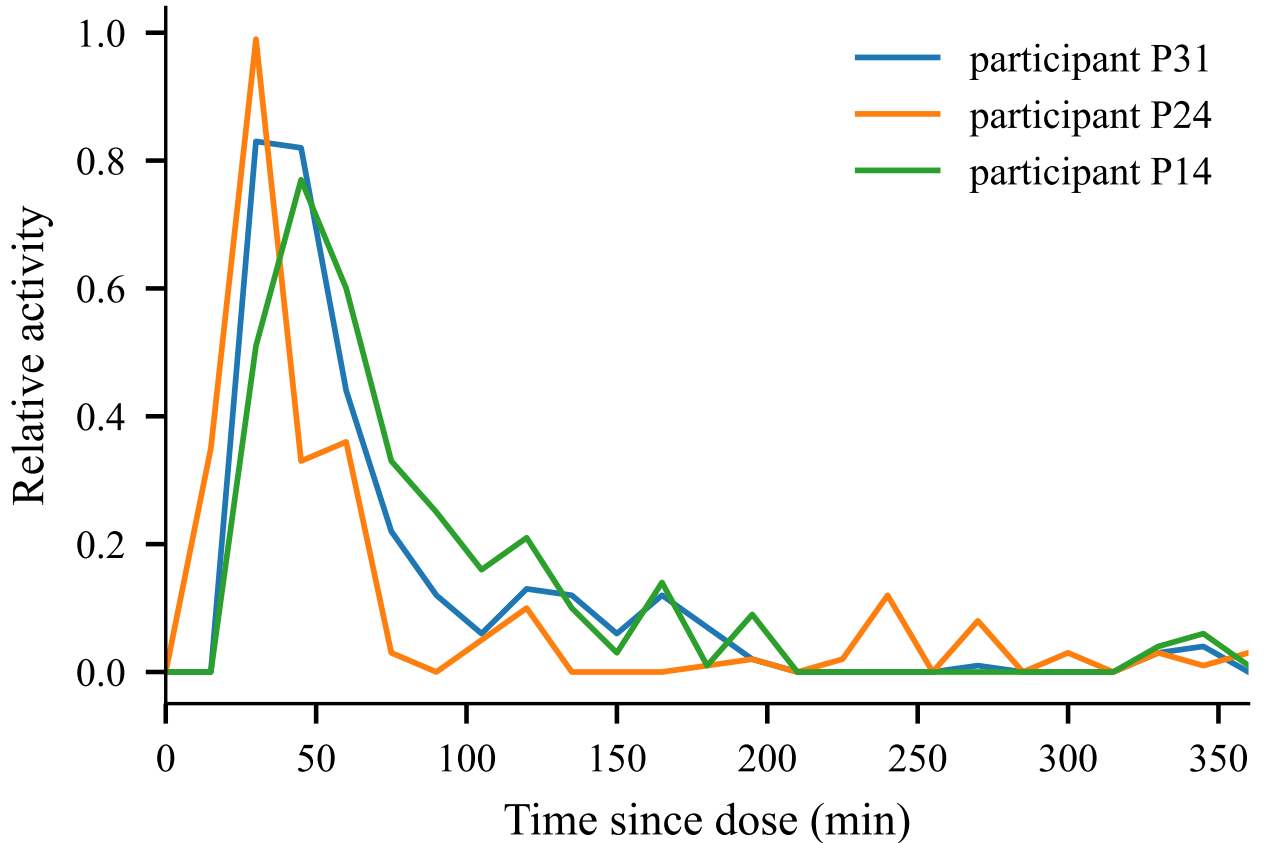


Figure 1. Estimated impulse responses for three representative participants. No functional form is imposed; the curve is the vector of lag coefficients from equation (5).

A peak was reported only where the estimated kernel possessed an identifiable mode (section 2.7). 2 participants (P1, P17) failed that test: the recovered curve was near-flat, with under a quarter of its mass within

half an hour of its maximum and a maximum that exceeded the curve three hours later by a few per cent. An argmax exists for any curve and could have been quoted, but it would describe noise rather than a response. Those participants are excluded from the peak statistics below, leaving 29 of 31.

Observed peaks had a median of 42 min (range 28 to 112). Configured values for the same participants had a median of 55 min (range 35 to 77). The correlation between configured and observed peaks was +0.03 (Figure 2). Configured values were sharply discrete, whereas observed values were distributed continuously (Figure 3).

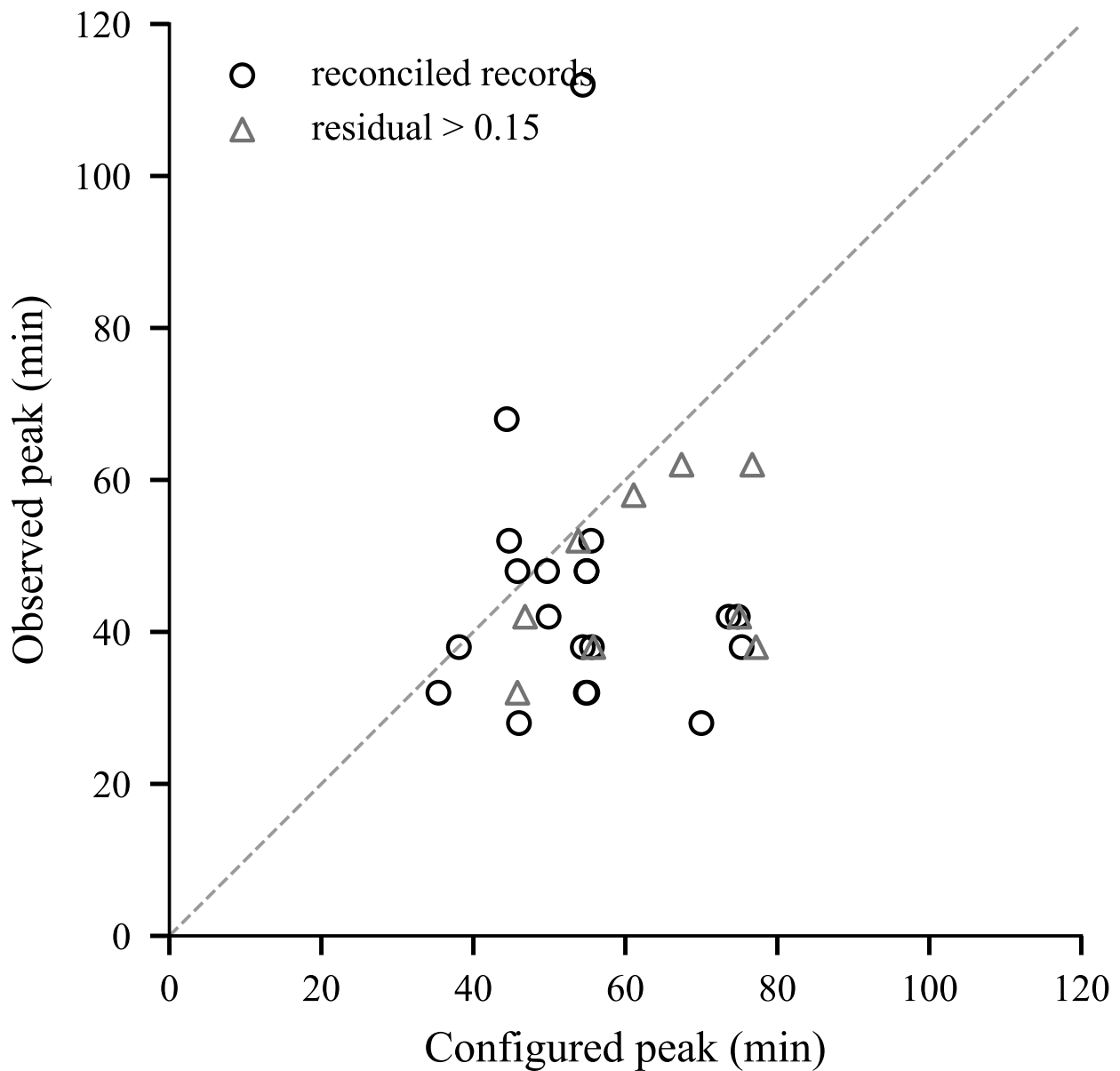


Figure 2. Observed against configured peak. The diagonal marks equality. Triangles denote participants whose dose records did not reconcile with their logged insulin-on-board (relative residual > 0.15).

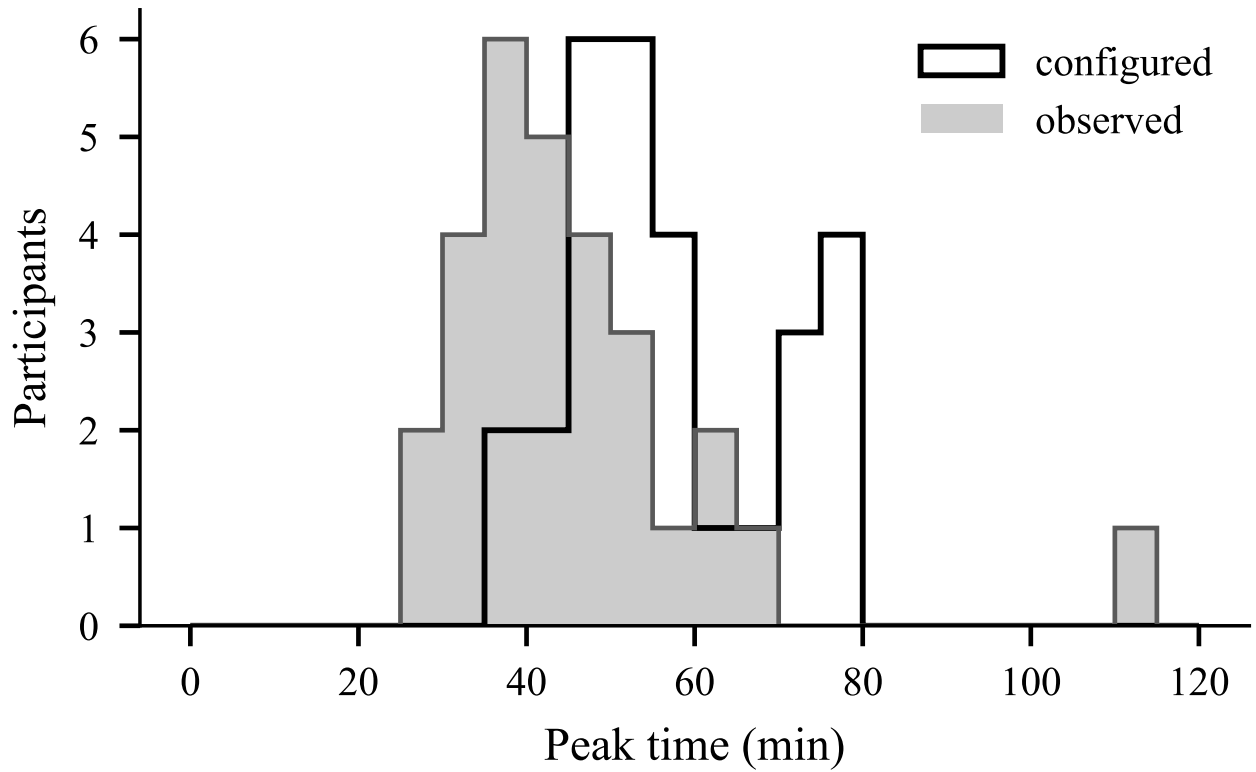


Figure 3. Distribution of configured and observed peak times.

The difference between observed and configured peaks had a median of -8 min; 13 of 29 participants agreed within 10 min and 8 differed by more than 20 min in the direction of a shorter observed peak. Uncorrected sensor lag biases observed values later [6, 7], so differences in this direction are conservative.

3.4 Duration of action

Duration of action was not identifiable in 10 of 31 participants under the leverage criterion. Separately, truncating each participant's fitted kernel at five hours altered the predicted insulin-on-board series by less than the fit residual in every participant, indicating that the tail carries no usable signal at these dose sizes.

3.5 Dose dependence

Doses were divided into bins of each participant's own dose distribution and one kernel fitted per bin, penalised for smoothness along lag and toward adjacent bins. Within individual participants this design was not identified: on simulated data in which every dose acted through a single 55-minute kernel, the recovered profile spanned 40 min rather than being flat. Pooled across participants the same control recovered the injected kernel to within 5 min, and the pooled estimate is reported on that basis.

Dose bin	Median dose (U)	Observed peak (min)	Control (min)
1	0.05	40	55
2	0.10	40	55
3	0.25	40	60
4	0.55	38	55
5	1.10	40	58

Table 3. Observed peak by dose bin, pooled across participants, against a control in which all doses share one kernel.

The observed peak was flat at 38 to 40 min across a twenty-two-fold range of dose size, with the control confirming that a tilt would have been detected had one been present.

3.6 Sensitivity analyses

Beyond the two corrected biases, the following were examined and found not to alter the estimates materially. The non-negativity constraint was binding at short lags, with three of the first six coefficients clipped to zero and substantial negative mass in an unconstrained refit, consistent with the controller dosing in response to rising glucose; removing the constraint changed the peak by less than 5 min, and suppression of the early kernel biases the peak later rather than earlier. Excluding days contributing fewer than twenty eligible samples changed no estimate. Requiring the successor sample also to satisfy the eligibility mask moved the peak by 0 to 5 min. Admitting net basal insulin as a second kernel moved the bolus peak by 0 to 5 min; the fitted basal kernel peaked at lag zero and carried 38–43% of total kernel mass, consistent with the controller's reaction rather than insulin action. Splitting records by glucose level, the peak was later in the high-glucose stratum in 11 of 16 participants (median +8 min).

Admitting insulin logged as delivered outside the pump (section 2.1) shifted the observed peak by a median of +0 min across the affected participants, with 2 shifting by 10 min or more. The effect is concentrated rather than general: it appears only where those records are too large to be single rapid-acting boluses, and is absent where they are of ordinary bolus size. The largest, in the participant whose external records reach 35 U at twelve-hourly spacing, moves the peak from 38 to 78 min and markedly reduces the prominence of the recovered mode — what a slower preparation represented by a rapid-acting kernel would do. The direction is not uniform: one further participant moves 10 min the other way, which is within the range this estimator moves under other specification choices of similar size.

4. Discussion

The configured profile is recoverable exactly from routine records. Because equation (4) is an identity rather than a model, the estimate is an audit of what a system is computing with, independent of what is reported elsewhere, and it identified both an undocumented change of profile during the observation period and, in one participant, a configured profile that did not correspond to the insulin they understood to be in use. In that case the recovered profile matched a default preset rather than the value associated with the insulin concerned; the alternative fitted the logged insulin-on-board several times worse, and the recovered value was stable across disjoint subperiods. Such records establish what a system is computing with; they do not establish why, and no claim as to cause is made.

The observed profile is recoverable with useful precision, but the estimate depends more on the identification strategy than on the choice of model family. The parametric within-window formulation was unbiased on its own generating model to within 0.3 min and shape misspecification biased it by only –10 to +5 min, yet its estimates varied over 40 to 90 min across defensible analytical choices. Pooling days to identify a clock-locked drift term resolved this. The intuitive alternative — allowing each window its own drift term — is what destroys identification.

Observed peaks were substantially shorter than both configured values and published clamp pharmacodynamics, and the correlation between configured and observed was +0.03. Three quantities are commonly treated as interchangeable that are not: the maximum glucose-lowering effect reported in product information (60–180 min, under clamp conditions at 7–30 U), the peak assumed by delivery systems (45–75 min, a convention rather than a label value), and the free-living response measured here (median 42 min, at 0.1–1 U with glucose free to fall). Four explanations for the discrepancy were examined. Estimator bias was excluded by per-participant injection of a known 75-minute curve, which returned 65 to 85 min. Omitted basal insulin was excluded. Counter-regulation, which a clamp suppresses by design and free-living conditions do not, contributed a median of +8 min — directionally as predicted but an order of magnitude too small. Dose dependence was excluded within the range these systems dose in, though the comparison with clamp doses spans two further orders of magnitude that these data cannot address.

Duration of action was not identifiable in the majority of participants, and its bootstrap interval was found to be narrowest precisely where the parameter was least identified. Any analysis reporting a duration with an interval, absent a leverage or stability check, risks reporting false precision.

The practical consequence of a configured profile slower than the observed response is that the controller carries insulin-on-board it believes still to act. Since insulin-on-board restrains dosing, this is a conservative error: at 60 min after a dose the median discrepancy was +0.058 U per unit delivered, though 13 of 31 participants exceeded 0.15 U per unit. 5 participants showed the opposite sign, in which the restraint is released earlier than the observed response supports. These figures quantify a disagreement within the controller's own accounting and do not measure an error in insulin action.

5. Limitations

The eligibility rule has a consequence for what the peak estimate can speak to. Target samples are excluded for three hours following any carbohydrate entry, so a dose given with announced carbohydrate cannot contribute to any lag below 180 min. Every lag in the region where the peak lies is therefore identified by insulin delivered without announced carbohydrate — corrections and automatic microboluses. Across participants the 90th-percentile dose supporting that region is a median of 80% of the 90th-percentile dose overall. This does not bias the peak in time, since a correction should follow the same kinetics as a meal bolus, but it does mean the dose-independence result covers the correction range rather than the full delivered range, and that the doses identifying the peak are precisely those given because glucose was high or rising.

1. Sensor lag, comprising interstitial delay of approximately 5 to 7 min [6, 7] together with filter delay, is not corrected. It biases observed values later and therefore cannot account for observed peaks shorter than configured.
2. Unannounced carbohydrate cannot be excluded, only rendered less likely. For participants who never enter carbohydrate the carbohydrate-on-board criterion is inert.
3. Physical activity is uncontrolled where the uploader provides no step count, which applies to all Trio participants.
4. The comparison with clamp pharmacodynamics spans doses two orders of magnitude apart and conditions that differ in whether counter-regulation is permitted; the quantities are not directly commensurable.
5. Estimates are observational. A difference between observed and configured values is a discrepancy and not evidence that altering a setting would improve glycaemic outcomes.
6. The sample is one of convenience, self-selected, and without demographic characterisation. One participant contributed only 16 days. No external replication has been undertaken.

6. Conclusions

An individual's configured insulin profile can be recovered exactly from their own AID records, providing an audit that detects profiles differing from those the user believes are in use and changes that no event record documents, without access to the device. The profile the glucose response follows can be recovered by non-parametric deconvolution provided the drift confounder is identified by pooling across days, and is robust to the controller feedback inherent in closed-loop data. It is systematically shorter than the configured value, and the configured value carries little information about it. Duration of action is not recoverable for most individuals. Resolving whether the difference reflects insulin behaviour or the conditions of measurement requires a designed study in which a bolus is given fasting with no other insulin.

Declarations

Ethical approval. This analysis constitutes secondary use of routinely collected data. The records were generated by participants' own devices in the course of their normal treatment and uploaded to personal Nightscout instances to which the author already had access; no data were collected for the purpose of this study, no intervention was made and no protocol was imposed. Formal ethical review was not sought on that basis. Participants are identified only by code, no demographic data were held, and no individual is characterised by any attribute other than the delivery system in use.

Data availability. Individual-level records cannot be shared. Analysis code is available at github.com/tim2000s/Insulin-Kinetics under AGPL-3.0. All numerical values in this manuscript are generated directly from the analysis outputs.

Funding. None.

Competing interests. The author is a contributor to open-source automated insulin delivery systems, including Boost, AndroidAPS and Trio, all of which are represented among the systems analysed here, and has an interest in the safety of all of them. No commercial interest is held in any of the systems or in any of the insulin products discussed.

Reproducibility. Generated 2026-08-07 from analysis outputs; anonymised participant labels.

References

1. Maksimović D. Exponential insulin activity curves. LoopKit/Loop issue #388, 2017. The functional form adopted by orefo and derivative systems; the curve shape and parameter values it interpolates derive from the clamp pharmacodynamics of references 8–12.
2. De Nicolao G, Sparacino G, Cobelli C. Nonparametric input estimation in physiological systems: problems, methods, and case studies. *Automatica* 1997;33(5).
3. Golub GH, Heath M, Wahba G. Generalized cross-validation as a method for choosing a good ridge parameter. *Technometrics* 1979;21(2):215–223.
4. Künsch HR. The jackknife and the bootstrap for general stationary observations. *Annals of Statistics* 1989;17(3):1217–1241.
5. Forssell U, Ljung L. Closed-loop identification revisited. *Automatica* 1999;35(7):1215–1241.
6. Basu A, Dube S, Slama M, et al. Time lag of glucose from intravascular to interstitial compartment in humans. *Diabetes* 2013;62(12):4083–4087.
7. Basu A, Dube S, Veettil S, et al. Time lag of glucose from intravascular to interstitial compartment in type 1 diabetes. *Journal of Diabetes Science and Technology* 2015;9(1).
8. Humalog (insulin lispro). EMA product information, section 5.1.
9. NovoRapid (insulin aspart). EMA product information, section 5.1.
10. Apidra (insulin glulisine). EMA product information, section 5.1.
11. Fiasp (faster-acting insulin aspart). EMA product information, section 5.1.
12. Lyumjev (insulin lispro-aabc). EMA product information, section 5.1.